

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int main(){
3      int i,n,high,low,mid,ans;
4      scanf("%d",&n);
5      int a[n];
6      for(i=0;i<n;i++)
7          scanf("%d",&a[i]);
8      low=0;
9      high=n-1;
10     ans=n;
11     while(low<=high){
12         mid=(low+high)/2;
13         if(a[mid]==0){
14             ans=mid;
15             high=mid-1;
16         }
17         else
18             low=mid+1;
19     }
20     printf("%d",n-ans);
```

21 | }

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓
✓	10 1 1 1 1 1 1 1 1 1 1 1	0	0	✓